

Mahesh Chitakoti

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SUMMARY

Backend & AI Engineer with **2.5+ years of experience** building production GenAI systems, backend microservices, and asynchronous pipelines. Specialized in **Python, FastAPI, TypeScript, RAG, vector databases (Qdrant), and LLM orchestration**, with a proven record of cutting LLM inference costs by **35%** and reducing high-traffic backend API latency by **60% (800ms → 320ms)**.

TECHNICAL SKILLS

Languages & Backend: Python, TypeScript, SQL, FastAPI, Node.js, NestJS, Express.js, AsyncIO, REST APIs, Microservices, WebSockets

GenAI & LLM Engineering: Retrieval-Augmented Generation (RAG), Agentic AI (LangChain, LangGraph), Tool Calling, Function Calling, Structured Outputs (Pydantic), Semantic Caching, Prompt Optimization

Vector Search & Retrieval: Qdrant, pgvector, Vector Similarity Search, Hybrid Search, Embeddings, Semantic Search, Reranking

LLM APIs & Cloud Infrastructure: Google Gemini API, OpenAI API, Anthropic Claude API, AWS Bedrock, AWS EC2, AWS S3, Docker, GitHub Actions CI/CD

Databases: PostgreSQL, Redis, MongoDB

EXPERIENCE

AI Engineer – SahioNext Technology Pvt. Ltd.

Oct 2025 – Apr 2026

Remote

- Built an asynchronous GenAI audio pipeline using **FastAPI, Google Gemini APIs, and Qdrant**, implementing **admission control (RPM/TPM tracking) and schema enforcement**, reducing inference costs by **35%**.
- Developed semantic-search microservices over lesson transcripts using **Qdrant vector similarity search**, indexing **10,000+ transcript chunks** to achieve **300ms retrieval** for automated CEFR proficiency analysis.
- Built WebSocket and MQTT streaming pipelines with exponential backoff and reconnection logic, processing **500+ daily asynchronous jobs** with zero message loss.
- Implemented operational telemetry dashboards in **React.js and Redux** tracking token consumption, API quotas, and pipeline stages, reducing mean time to resolution (MTTR) by **70%**.

Full Stack Developer (Contract) – PearlThoughts

Feb 2025 – Sep 2025

Remote

- Engineered automated backend integrations using **NestJS, TypeScript, and Google Workspace APIs**, executing over **1K automated workflows/day** across multiple client engagements.
- Built distributed workflow pipelines with **exponential backoff, idempotent webhooks, and rate limiting**, achieving **99.8% job execution reliability** across multi-tenant clients.
- Containerized backend services using **Docker** and deployed them on **AWS** through automated **GitHub Actions CI/CD** pipelines, standardizing deployments across environments.
- Validated coding labs, grading infrastructure, and deployment workflows for **Coursera course launches**, identifying release blockers before production deployment.

Full Stack Developer – The DevOps Team

Oct 2023 – Jan 2025

Bangalore, India

- Optimized high-traffic **Node.js and Express.js APIs** using **Redis caching and database query optimization**, reducing median response time from **800ms to 320ms**.
- Developed job recommendation workflows that matched candidate profiles against job requirements using multi-attribute filtering to surface more relevant roles.
- Built **React.js and Next.js** onboarding experiences with integrated **A/B testing**, lifting user signup conversion rates by **18%**.
- Implemented real-time notification systems using **WebSockets** and supported **Docker- and Kubernetes-based** production deployments in collaboration with DevOps teams.

PROJECTS

WorkBridge – AI-Powered Collaboration Platform — Personal Project

May 2026 – Present

React.js, Next.js, Node.js, Express.js, Python, PostgreSQL, pgvector, Redis, OAuth | Live Project

- Built a full-stack collaboration platform connecting students with industry projects through skill-based matching and real-time collaboration workflows.
- Built REST APIs using **Node.js, Express.js, PostgreSQL, and Redis**, implementing role-based access control, OAuth, and response caching to achieve **150ms p95 latency**.
- Developed an AI recommendation service using **Python, pgvector, and cosine similarity matching**, pairing student skill profiles to project requirements with sub-second ranking retrieval.

CERTIFICATIONS

Oracle Agentic AI Certified Foundations Associate	– Oracle	Aug 2026
Oracle AI Vector Search Certified Professional	– Oracle	Sep 2025
Machine Learning I	– Columbia University	Aug 2025
Oracle Cloud Infrastructure 2024 Generative AI Certified Professional	– Oracle	Aug 2024

EDUCATION

Adichunchanagiri Institute of Technology	Chikkamagaluru, KA
<i>Bachelor of Engineering, Visvesvaraya Technological University (VTU)</i>	<i>Aug 2023</i>